

2.3 Preprocessing

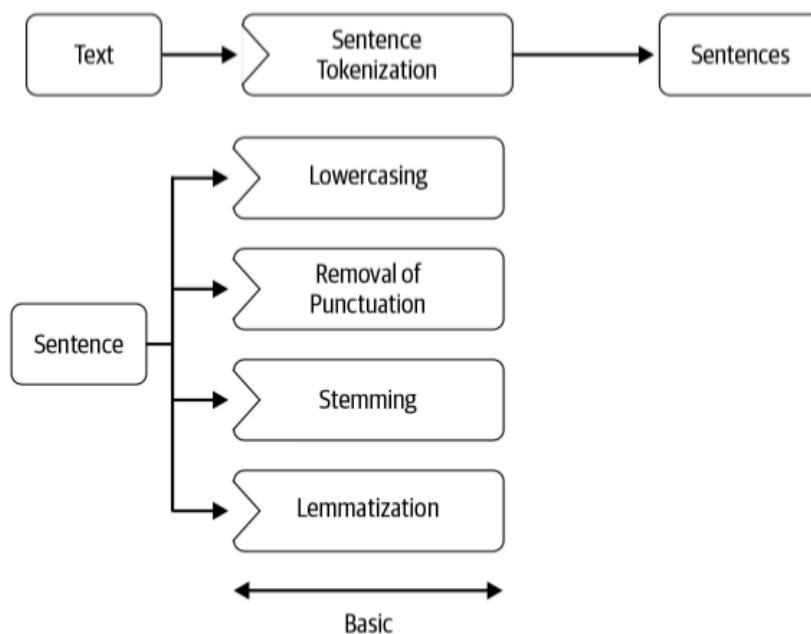
Tokenize, normalize, drop stops, and choose stemming or lemmatization.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. Canonical form

Cleanup gave you a string. **Pre-processing** turns that string into a sequence you can count, stem, or embed.

Common steps:



Not every task wants every step. Named-entity work wants capitalization. Sentiment often wants **!** and emoji. A search index may want stems. Decide from the job, then keep the recipe fixed so train and test match.

2. Tokens and types

A **token** is one occurrence. A **type** is the distinct string.

To be or not to be → 6 tokens (to, be, or, not, to, be) and 4 types (to, be, or, not). The set of types is the **vocabulary**.

Whitespace split is the baseline you already know:

```
"Every time I learn something new, it pushes some old stuff out of my brain.".split(" ")
```

You get `new`, and `brain.` glued to punctuation. `str.split()` is not a tokenizer.

Use `nltk.word_tokenize` or `spaCy`. They split words *and* punctuation. That is what Lab 1 already showed.

3. Punctuation, case, stop words

Do not strip all punctuation by reflex.

Why keep it	Example
Boundaries	<code>.</code> <code>?</code> <code>;</code> mark sentences and clauses
Illocution	<code>!</code> and <code>?</code> change the act of the sentence
Sentiment	<code>: -)</code> and emoji are features

Lowercasing merges `Apple` the company with `apple` the fruit. That is good for topic bags of words. It is bad for names.

Stop words (`the`, `a`, `of`, `is`) are high-frequency function words. Dropping them shrinks a count vector. Do not drop them for tasks that need grammar or polarity (`not` is a stop word in many lists and it flips sentiment).

```
from nltk.corpus import stopwords
stops = set(stopwords.words("english"))
[t for t in tokens if t.lower() not in stops]
```

4. Stemming vs lemmatization

English (and many other languages) inflects: `organizes`, `organized`, `organizing`. You often want one index term.

	Stemming	Lemmatization
Method	Chop a suffix with a heuristic	Morphological analysis → dictionary form
Example	<code>organizes</code> → <code>organ</code>	<code>organizes</code> → <code>organize</code>
Output	May not be a word (<code>flying</code> → <code>fli</code>)	Aims at a real lemma
Tool	<code>PorterStemmer</code>	<code>WordNetLemmatizer</code> (needs POS for best results)

Stemming is fast and crude. Lemmatization is slower and usually cleaner. Lab 1 already ran both on the same list. Use one consistently; do not mix them in the same vocabulary.

5. Other normalization

Typical extra steps:

- map digits (9 → nine , or all numbers to #)
- expand a small set of abbreviations (Dr. → Doctor) if the domain is stable
- collapse repeated characters (soooo → soo) on social text

Language detection belongs near the front. Most tokenizers, stop lists, and stemmers are language-specific. English tools on Spanish (or on a mix) produce garbage, not a slightly worse model.

Code mixing is two or more languages in one post. Tokenizers trained on one language break. Language ID on the whole document is not enough.

The diagram shows the words of a sentence with brackets underneath each word indicating its language. The words and their language labels are: Dey. (Tamil), wǒ men (Mandarin (我们)), paktor (Cantonese (拍拖)), always (English), makan (Malay), at (English), kopitiam (Malay), Hokkien/Hakka (店) (Hokkien/Hakka), and one. (???).

Translation: Hey, when we date we always eat at the coffeeshop (one).

6. A short recipe

For a bag-of-words classifier on English reviews, a default that is good enough:

1. language-filter to English
2. tokenize (not split)
3. lowercase
4. drop stop words *except* negators if you care about sentiment
5. lemmatize (or stem, not both)

For POS or NER (note **2.4**), skip lowercasing and stop-word removal.

7. Practice

1. Tokenize Wow, we can finally stop using the split function! by hand the way NLTK would. How many tokens?
2. Give one task where you should *not* lowercase, and one where you should.
3. Why can flying → fli still be useful in a search index even though fli is not a word?